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Comparing Arm Tracking Teleoperation to Conventional Joystick Control of a Laterally Mounted Robot Arm

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Abstract

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1 Introduction

The goal of this project is to create a system for controlling and teleoperating a robot arm with a human arm. In order to maintain simplicity in the project scope, the robot arm is assumed to be shoulder-mounted and has the seven degrees of freedom (DoF) [1], matching that of a healthy human arm. An example of the humanoid robot arm (the YuMi robot) is shown in figure 1.

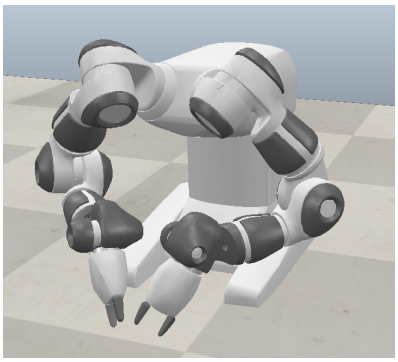


Fig. 1. YuMi Robot Arm To Be Used In Experiment

1.1 Background and motivation

The motivation for this project comes from articles within the books: “Human–robot teleoperation based on Imus” [2] and “Whole-body human-motion based robot teleoperation” [3]. Both of these articles use an inertial measurment unit (IMU) to gather data, but only maps the wrist position to the robot’s end effector.

Redundancy in the degrees of freedom of the arm means that a valid end effector position can have multiple positions of the previous links [4]. Therefore this ambiguity could lead to collisions with obstacles in the environment, and lack of safety for people within the envelope of the robot arm.

Some specific use cases will require absolute control of every link of the robot arm - for example underwater sample collection. Whilst Jin et al. describes a potential autonomous solution [5], this method is comparatively slow to the prospect of remote teleoperation. Consequently, the complexity increase for the case where nothing is stationary, a more realistic use case, would mean that this method would be inefficient compared to the intuition of a trained operator.

Additionally, Jankowski and Grabowski shows that for tasks involving teleoperation of a robot for pick and place tasks, virtual reality style mapping of head movements with arm to end effector

mapping is more efficient and less error prone than a typical joystick [6]. Hence, exploration of the performance characteristics between conventional control methods and arm mapped teleoperation is undertaken.

1.2 Aims and objectives

The overall aim is to compare the performance of a system for mapping the joint angles of a human arm to a shoulder-mounted robot arm to conventional control methods. This will include:

- Identifying and comparing existing technologies, including IMU based solutions discussed earlier, attitude and heading reference systems [7], computer vision based systems [8], and flex sensor based systems [9].
- Developing a version of the tracking system suitable for controlling a simulation of the YuMi robot arm.
- Devising experiments appropriate to comparing conventional methods of control to arm tracking teleoperation methods.
- Analysing the difference, if any, in performance between either method of control in a variety of environments.

2 Examination of existing human to robot arm teleoperation methods

This section explores which arm tracking methods are the most suitable for this project. Quantitative and qualitative data from secondary research is used to create a justified specification for the final solution. Finally, a Pugh matrix will be applied to fairly score and select an appropriate method to further develop.

2.1 Why choose teleoperation over conventional input methods

The measurement of human arm joint angles has been an interest of academia for several decades: with one of the first examples being demonstrated by Liu et al. In industrial settings, full body tracking has primarily been used in the film industry to accurately measure an actor's performance [11] and map them onto computer generated characters [12].

This approach could also be applied to the control of robotics: have a skilled operator remotely control industrial robots where traditional control methods (e.g. joysticks) are unsuitable. Research has shown that this approach could lead to 1.8 times faster task completion time and with a quarter of the errors (1.54% compared to 8.65%) in the best cases [6].

Another way to increase the performance of a task with teleoperation is mapping the head movements of an operator to the camera of the robot performing the task [13]. However, this study was based primarily on the control of wheeled robots; manual joystick controls exceeded the performance of teleoperation in this specific task.

These two cases suggests that whether teleoperation is more effective than traditional joysticks is strongly dependent on the task. Overall, in the case of robot arm control, there is potential for performance increases via human arm teleoperation [2], strengthening the justification for this project.

2.2 Requirements of a suitable tracking system

In order to compare the final system to existing systems and research, desired characteristics must be identified.

Accuracy requirements - Be able to measure joint rotations with a minimum resolution of $3°$.

Experienced surgeons have been measured to have $5°$ of vertical movement range on all joints when asked to hold their hand on a target, and close to $3°$ range for horizontal movement [14]. Therefore any measurement lower than this resolution can be characterised as 'noise' from human inaccuracy, and can be smoothed out.

Latency requirements - Have a maximum system latency of 300 ms.

Teleoperation task performance generally degrades around the two key figures of 600 ms and 1500 ms [15]. Thus, the Nyquist criterion equivalent of this threshold is used as a well informed estimate of how much latency can be present in the system, leaving a final requirement of 300 ms.

Resource requirements - The budget for this project is £180, and 30 unit credits worth of time, equating to 240 hours of time commitment.

Both of these factors will mean that the final solution favours simpler, more complete solutions. Anything novel is a potential failure point that could mean the project does not complete.

2.3 Survey of existing tracking technologies

In order to fully evaluate the myriad of arm tracking methods available, a Pugh Matrix has been created (see Table 1) to evaluate and compare several factors between each method. Each method will score $1 - 5$ in each category, with a brief comment justifying the score in each column.

Below are the decision factors of the Pugh Matrix and a brief description of how they are scored:

Complexity - This is calculated based on a well reasoned estimate of the total time to build the system. Each multiple of 48 hours reduces the score by 1, e.g. $0 - 48$ hours would score 5, $49 - 96$ hours would score 4.

Cheapness - This is calculated based on a well reasoned estimate of the total cost of building the system. Each multiple of £36 reduces the score by 1, e.g. £$0 -$ £36 would score 5, £$37 -$ £72 would score 4.

Accuracy - This is calculated from available research on the accuracy of this method. Each multiple of $5°$ above $5°$ reduces the score by 1, e.g. an absolute accuracy of $6 - 10°$ would score 4, $11 - 15°$ would score 3.

Latency - This is calculated from available research on the time it takes to obtain a single measurement of all 7 joint angles from this method. Each multiple of 100 ms above 300 ms reduces the score by 1, e.g. a latency of 301 ms $- 400$ ms would score 4.

Analysis of existing methods shows that the two computer vision approaches score similarly in terms of viability, therefore further analysis will be required to confirm the viability of each method. These will be performed in the form of accuracy and latency experiments. This has the advantage of being able to examine more factors than existing literature - of particular interest is the accuracy of measuring multiple joints angles simultaneously.

In particular, the potential accuracy advantage of the marker tracking method makes it more appealing, though this is offset by the stupendous cost of conventional Vicon camera systems. However, open source software along with ArUco markers could be used to offset this cost - which is

worth exploring before committing to any substantial development. For example, Wieczorek et al. describes using ArUco markers along with standard home cameras to perform specialised kinematic analysis [16].

Table 1. Pugh Matrix with detailed scoring justifications

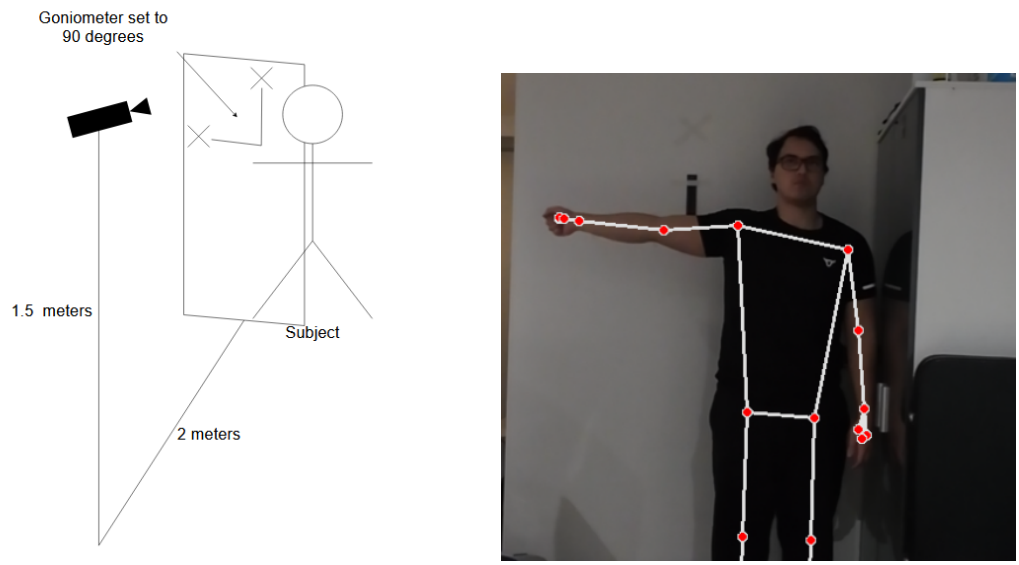
Method	**Complexity** $1-5$	**Cheapness** $1-5$	**Accuracy** $1-5$	**Latency** $(1-5)$	**Total**
Pose estimation via computer vision	5 Existing solutions reduce complexity $30-40$ hours	5 £30 USB Camera	3 $10^\circ - 15^\circ$ [17]	5 ≤ 100 ms [18]	18
Marker tracking via computer vision	5 Very complex, though can be simplified through FOSS	3 £180+ for Vicon, cheaper solutions exist	5 <5° error	5 100-200ms	18
IMU-based	3 100 hours including PCB design + drift correction	4 £$50-$ £60	3 2.8° [19]	5 <1 ms [20]	15
Flex Sensors	2 150 hours	4 £40-60	4 10° [21]	5 ≤ 100 ms	15

3 Comparing two viable computer vision based methods of robot arm teleoperation

The criteria selected both ArUco marker tracking and pose estimation as equally viable methods of human arm tracking. This section aims to develop a minimum viable product (MVP) version of both methods to be used to quantify which approach is more suitable for the final solution.

3.1 Comparison of both computer vision methods

In order to evaluate the accuracy of both computer vision methods, a simple experiment was set up to measure the absolute accuracy on the joint angles that can be measured. A goniometer is attached to the wall at $90°$, with two end positions marked at $90°$ and $180°$. The subject will move their arm repeatedly between these markers, and the recording is captured by a camera as described in Fig. 2.



(a) A diagram of the experimental setup. **(b)** Example data capture

Fig. 2. The experimental setup, showing data capture.

The subject will then flex and extend their elbow between the two endpoints. In order to obtain a sufficient amount of data, 5 videos are recorded, each showing 10 flexion and extensions.

This creates enough samples to perform ANOVA between each video, to maintain experimental integrity. In order to measure error, the graph of elbow flexion angles is extended towards the $90°$ and $180°$ endpoints. This secondary graph is then subtracted from the first graph to obtain the error value, from which we can calculate the root mean square error, and is shown in Fig. 3.

Upon analysis of the RMSE, it was found that this value was $25.83°$. which was greater than the literature value of $10.37°$ [17]. Frame by frame analysis of the video showed that this descrepancy was not attributable to human error - the arm was clearly flexing and extending to the correct endpoints.

Plotting the inferred joint positions frame by frame revealed the culprit - perspective! The machine learning model used for joint position extraction is finding extra movement in the depth axis of the

Fig. 3. Showing how RMSE is calculated from elbow flexion data.

video, and attributing it to roll in the elbow axis, instead of flexion. This is clearly shown in Fig. 4.

To analyse the performance of joint angle tracking in the three dimensional case, a simple movement path is created. A 7 DoF manipulator is created from relevant DH parameters (specifics found in appendix Table **??**) These movements are then fed through a forward kinematics processing program to obtain a visualisation of this movement path, and this movement is recreated by the subject.

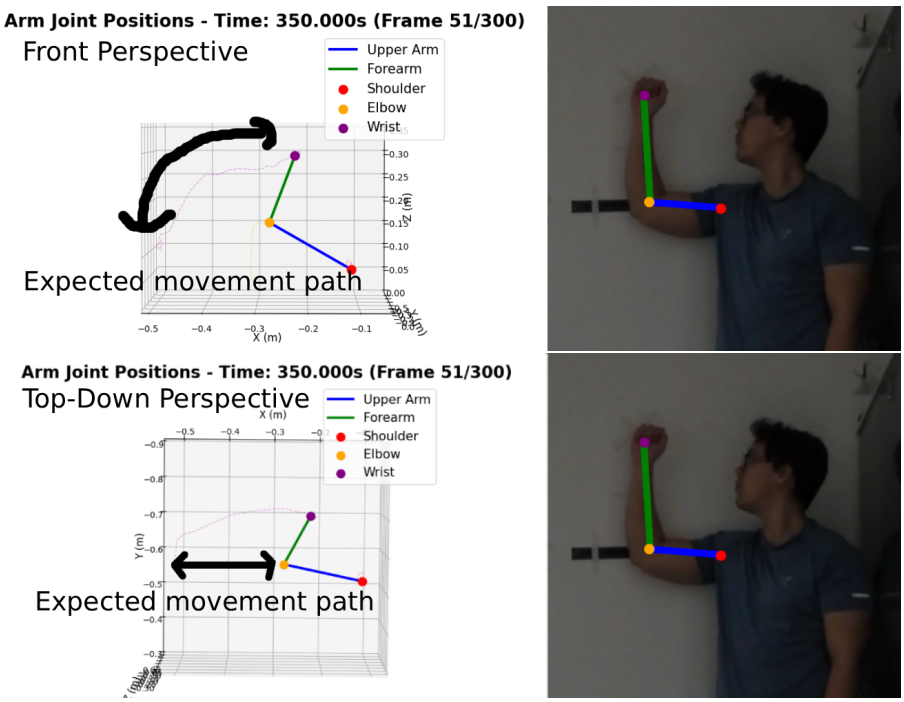


Fig. 4. Front and top view of inferred 3D joint positions.

This path is shown in 5. This movement has been selected to be easily recreated, whilst showing movement in three different joint angles simultaneously. The same data processing for the two dimensional elbow flex case is used to obtain RMSE of three different joint angle movements.

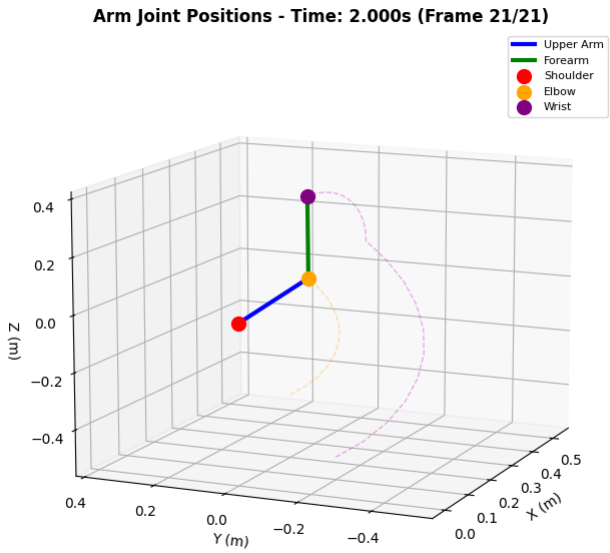


Fig. 5. 3D Joint Motion Path

3.2 Results of joint angle tracking methods

Table **??** shows the compiled RMSE errors for tracking each joint in both the two dimension elbow flexing case and the three dimension movement path.

Table 2. RMSE error of either tracking method.

Method	**RMSE Error Elbow Flexion 2D (°)**	**RMSE Error Elbow Flexion 3D (°)**	**RMSE Error Shoulder Flexion (°)**	**RMSE Error Shoulder Abduction (°)**
Pose estimation	25.83	7.19	6.89	11.68
Marker Tracking	N/A	N/A	N/A	N/A

Whilst the two dimensional RMSE error is larger than the literature suggests, the three dimensional joint error shows better accuracy than expected. Apart from the shoulder adduction/abduction measurements, each joint being tracked is almost in the same order of magnitude of the 3° requirements of the project.

Despite various attempts, ArUco marker tracking seemed to be too unstable in this specific application. Both 2D paper markers and 3D printed markers were used, though the glossy surface fin-

ish of either meant that tracking was unreliable at close distances and impossible at moderate distances. This was obtained from an experiment wherein a marker is held up to the camera and rotated around various axes, measuring the captured rotation.

Fig. 6 shows a plot of the measured roll angle, and shows the gaps in which no measurement was made. Likewise, the background conditions of the image also made a major difference to whether or not the marker was tracked. For example, Fig. 7 shows the marker not tracking whilst not distorted by angle at all, compared to being tracked at a very large angle. Clearly this level of unreliability is unsuitable for the final tracking task, and therefore no further testing will be required.

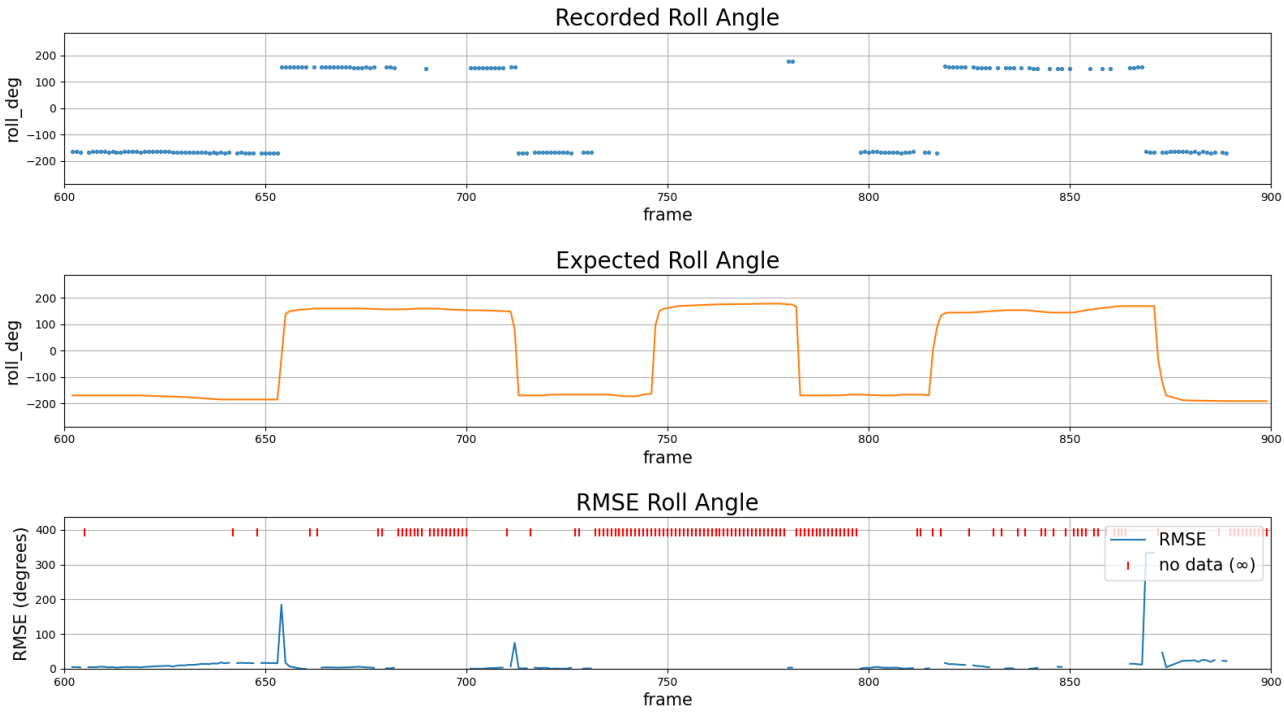


Fig. 6. Measured vs Expected graph of roll angle of ArUco marker.

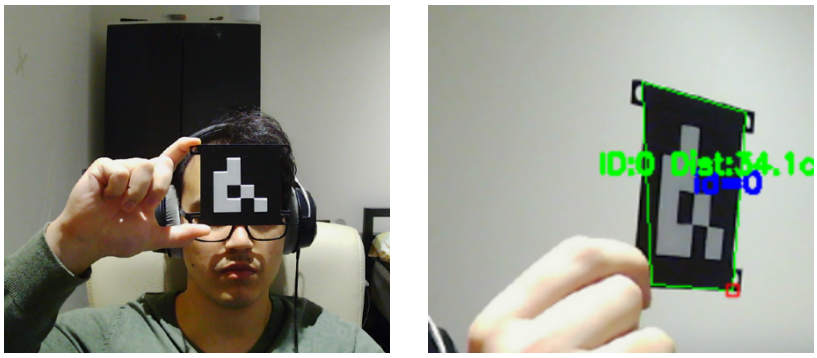


Fig. 7. Background dependence of the ArUco marker.

An interesting aspect that was encountered when performing the accuracy characterisation experiments with pose estimation is shown in Fig. 8, and points to a downside not immediately apparent. The program uses machine learning to create an estimation of joint locations in an image and was

ostensibly trained on a large corpus of human images. Hence, there was trouble distinguishing between real humans and pictures of humans being worn on the subject's clothing - a downside in the favour of the marker tracking method.

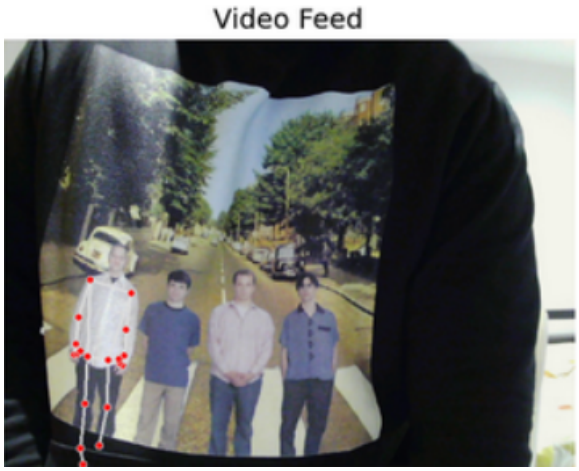


Fig. 8. Pose estimation software mistaking a picture of humans printed on clothing for an actual person.

3.3 Final Tracking Method

Results show that the pose estimation model is not able to achieve the $3°$ required for the project, it is able to get within the same order of magnitude of accuracy. Moreover, the mediapipe model used to power this is optimised for real time measurement; using a 30 fps camera, each measurement takes 33 ms, well below the required 300 ms. Hence, the comparison between conventional joystick control and human arm tracking teleoperation will use the pose estimation approach for comparison.

4 Arm tracking vs conventional control experiments

This section explains three scenarios that have been devised in CoppeliaSim to try and characterise the performance of each tracking method.

4.1 Description of experiment to compare conventional vs human arm tracking

In order to compare fairly the two methods of control, three scenarios will be examined:

Section 4.2 - Position matching task similar to Nisky et al. [14]. A random location within the envelope of the arm is selected and the operator must control the arm to this location as fast as possible.

Section 4.3 - Pick and place task. An object and target location spawns within the envelope of the arm, and the operator must manipulate the object to the target.

Section 4.4 - Obstacle laden pick and place task. Similar to pick and place task, but with tracking of obstacle collisions.

The operator must stand further away from the screen to be in frame of the camera for the pose estimation method. To negate the perception advantages this would incur for the controller tasks, the controller method will be performed sitting this same distance away.

Scenarios 2 and 3 consist of four different phases shown in Fig. 9 The object is considered 'held' as long as YuMi's grippers are within a certain distance threshold, and transition from an open state to a closed state. This prevents the experiment from being limited by the design and simulation of a suitable end effector - though this may act as an interesting extension for future research.

To measure the performance of the position matching task, the distance covered is divided by the time taken to find the normalised speed measurement. To measure the performance of the pick and place task, the cumulative time spent in each phase is measured, along with the overall time to complete the task. Likewise, to ensure fairness between each method in both pick and place tasks, seeded randomness will be used to ensure that both methods get the same trial spawn position.

4.2 Scenario 1: Random position matching task

100 samples were taken, with 10 runs of 10 experiments. Analysis of variance (ANOVA) shows no significant deviation between each run (see appendix for detailed ANOVA results). The controller method was roughly 42% faster than the pose estimation algorithm, although with a lower standard deviation of 0.043 compared to 0.053.

The speed difference is likely attributable to the operator's familiarity of using controllers for other tasks. Notably, the difference in standard deviation could imply that pose estimation is stronger for tasks that require consistency, rather than outright speed.

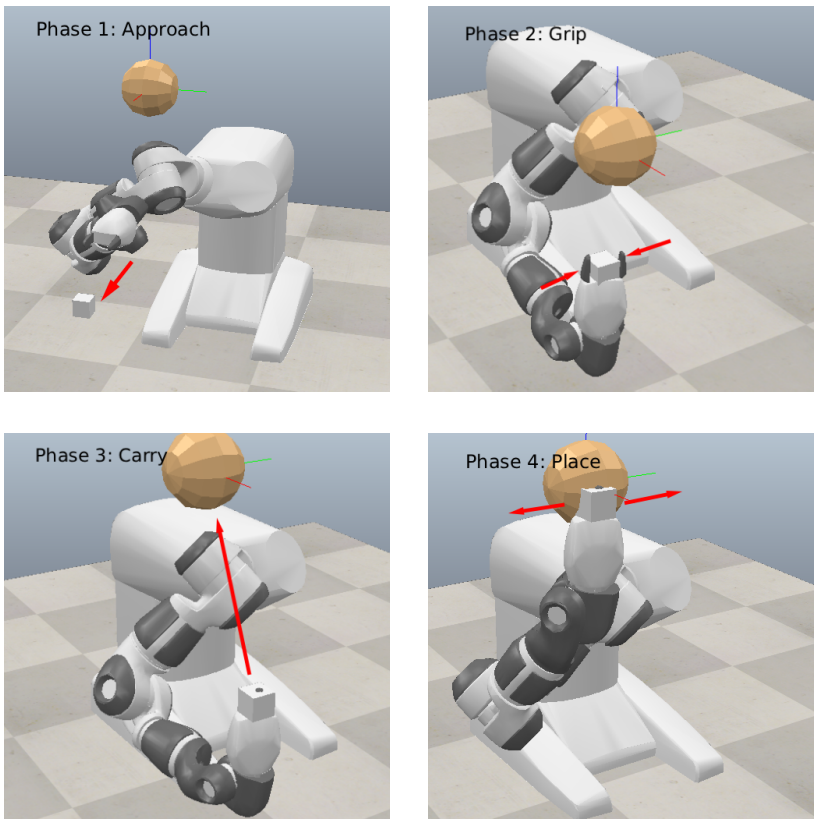


Fig. 9. Phases of the pick and place scenario

4.3 Scenario 2: Simple environment pick and place task

The pose estimation algorithm shows 60.3% reduction in time over the controller control in this task, with the noticeable difference being in the proportion of time spent in the 'approach' phase (see Fig. 11) The controller method requires the operator to orient the gripper over the object, and doing this in certain arm positions can lead to singularities in the inverse kinematics solver, hence the longer time.

These singularities can be seen in a joint angle over time plot of one of the experiments in Fig. 12. The advantage of the pose estimation method is that the human arm of the operator acts as a safety net for unreachable positions.

4.4 Scenario 3: Obstacle-rich environment pick and place task

The pose estimation algorithm is quicker on average by 57.2% in this task, likely for the same reasons as above - where the controller solution is running into singularities. Additionally, the added dexterity shown led to 3.46 fewer collisions (4.14 vs 7.60) on average compared to the controller

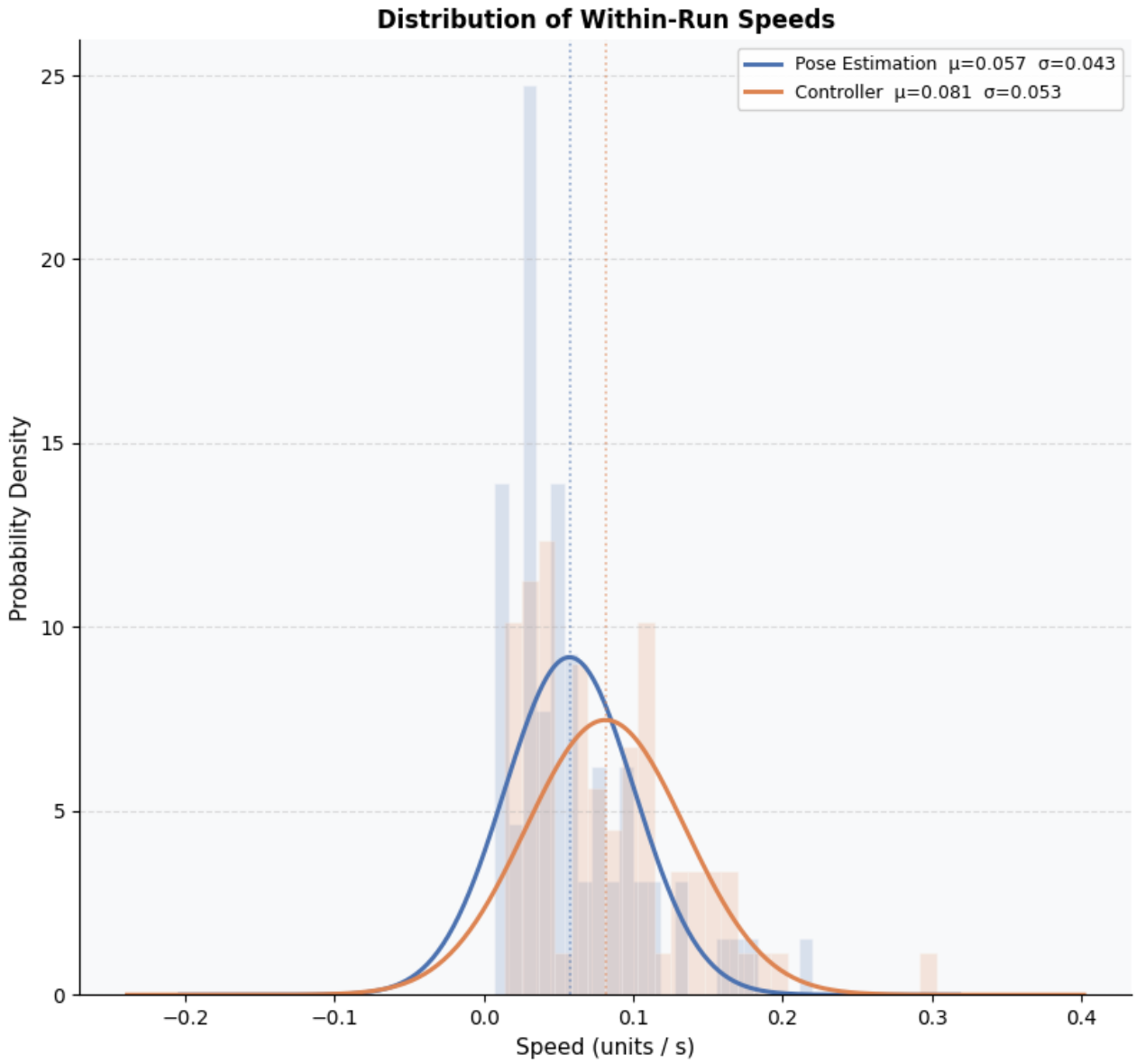


Fig. 10. Standard distribution plot of the results of either experiment

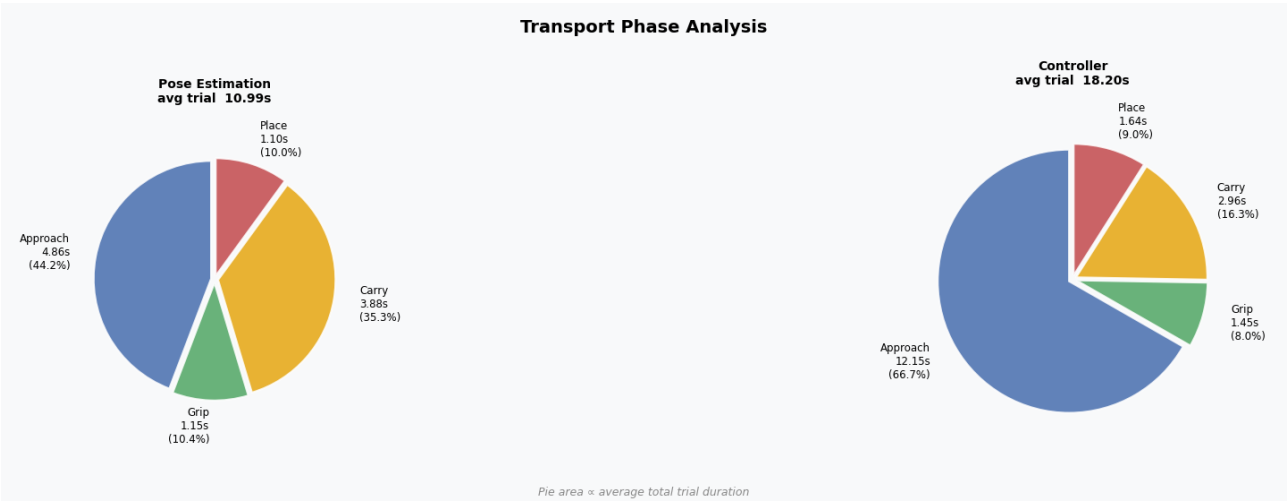


Fig. 11. Standard distribution plot of the results of either experiment

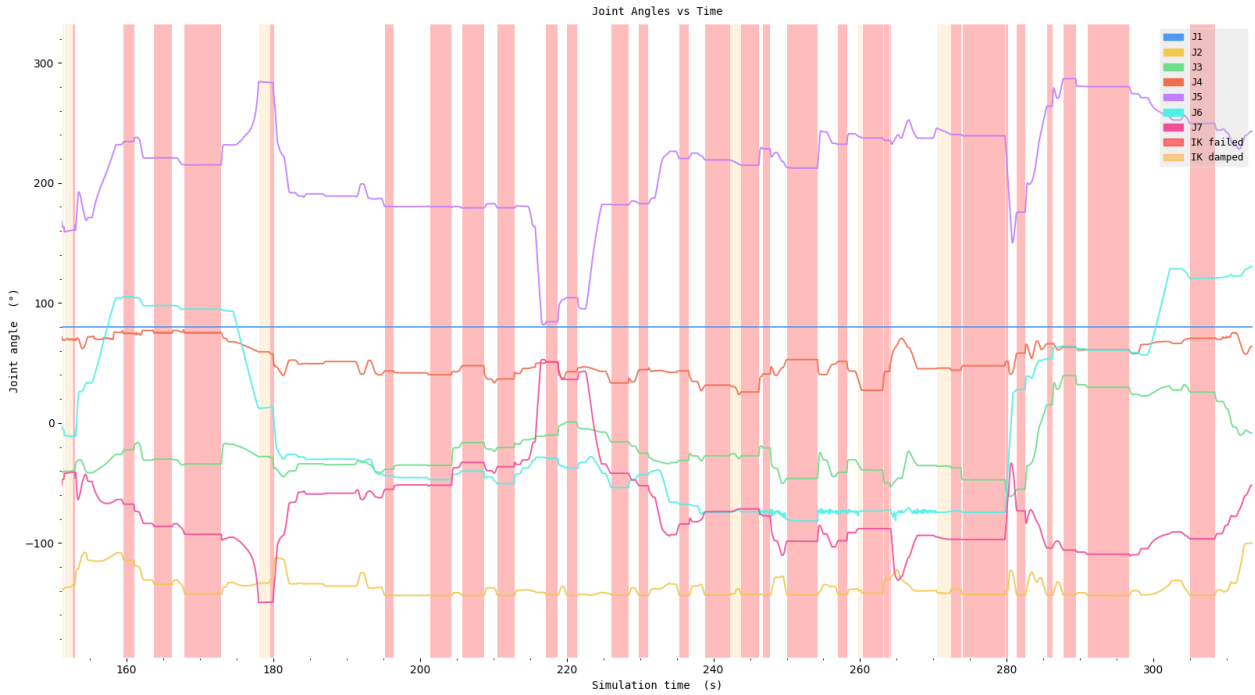


Fig. 12. Joint angle over time plot of the robot arm during a controller experiment

solution.

The fewer collisions may be due to more control that the operator has over the entire arm, as more data can be transmitted from the operator to the robot, in a way that is intuitive. Likewise, the smooth control of the pose estimation algorithm may make it more predictable compared to the IK approach of the controller solution - where it may be jumping over singularity positions unpredictably.

The proportion of time spent in the approach phase increased by 9% for the pose estimation solution, and decreased by 66.8% for the controller solution. Interestingly, the controller solution becomes more balanced in the time spent in each movement phase, whereas the pose estimation solution becomes front loaded.

There is also a noticeable difference in the average trial time between scenario 3 and scenario 2, likely attributable to the operator becoming more skilled and experience with each control method and experiment.

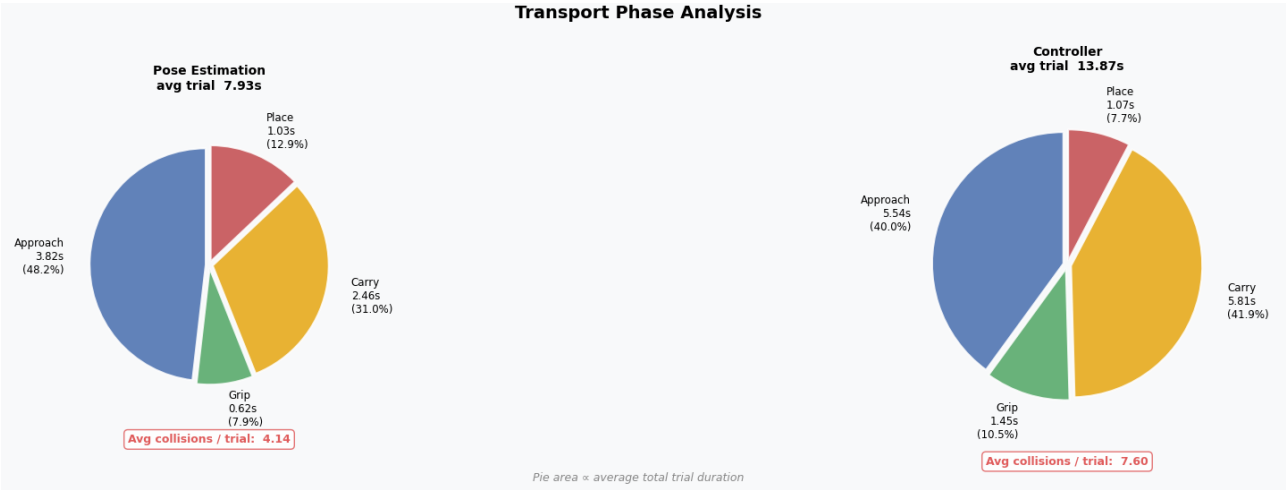


Fig. 13. Standard distribution plot of the results of either experiment

5 Conclusions and future work

5.1 Conclusions

5.2 Future work

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Appendices

A Project outline

Project outline as submitted at the start of the project is a required appendix. Put here.

B Risk assessment

Risk assessment is a required appendix. Put here.

C DH Parameters

The DH parameters used for the arm are as follows:

Joint	Description	θ_i	α_i	d_i	a_i
1	Shoulder abduction/adduction	q_1	$\frac{\pi}{2}$	0	0
2	Shoulder flexion/extension	q_2	$\frac{\pi}{2}$	0	0
3	Shoulder rotation	q_3	$-\frac{\pi}{2}$	0	L_u
4	Elbow flexion/extension	q_4	$\frac{\pi}{2}$	0	0
5	Forearm rotation	q_5	$-\frac{\pi}{2}$	0	L_l
6	Wrist flexion/extension	q_6	$\frac{\pi}{2}$	0	0
7	Wrist radial/ulnar deviation	q_7	0	0	0

Table 3. Denavit-Hartenberg Parameters for 7-DoF Arm (where L_u is upper arm length and L_l is lower arm length)

D ANOVA Figures